

Validation against held-out adjudicated labels

source: outputs/runs/run-20260922T045731Z/module_b/validation_report.json (research edition)

```
{
  "module_a": {
    "truth_pairs": 95,
    "predicted_pairs": 123,
    "true_positive": 94,
    "false_positive": 29,
    "false_negative": 1,
    "precision": 0.7642,
    "recall": 0.9895,
    "refusal_ratio": 0.0956
  },
  "module_b": {
    "snapshots": 1037,
    "test_snapshots": 208,
    "outcomes": {
      "payment": 539,
      "still_open": 488,
      "censored": 2,
      "close_no_payment": 8
    },
    "models": {
      "B0_age_stage_baseline": {
        "status": "FITTED",
        "snap_brier": 0.5144259569702185,
        "snap_log_loss": 0.7357284660871263,
        "n_snapshots": 206,
        "n_abstain": 0
      },
      "B1_discrete_multinomial": {
        "status": "FITTED",
        "snap_brier": 0.4618050188966311,
        "snap_log_loss": 0.6692132603100102,
        "n_snapshots": 206,
        "n_abstain": 0
      },
      "C1_quality_augmented": {
        "status": "FITTED",
        "snap_brier": 0.4615831802291266,
        "snap_log_loss": 0.6682848894134452,
        "n_snapshots": 206,
        "n_abstain": 0
      }
    }
  },
  "ablations": {
    "quality_features": {
      "B1_snap_log_loss": 0.6692132603100102,
      "C1_snap_log_loss": 0.6682848894134452,
      "delta": -0.0009,
      "interpretation": "quality features add no incremental value on the locked test set"
    },
    "candidate_vs_baseline": {
```

```
    "B0_snap_log_loss": 0.7357284660871263,  
    "B1_snap_log_loss": 0.6692132603100102,  
    "relative_improvement": 0.0904  
  }  
},  
"limitations": {  
  "study_scope": "All figures come from the study dataset of three participating sites.  
No research claim transfers to real operations.",  
  "censored_share": 2,  
  "close_class_rarity": 8,  
  "leakage_audit": "T10/T13/T14/T15 enforced: as-of truncation, forbidden fields,  
lineage-grouped splits, train-only preprocessing."  
}  
}
```